Lecture 8: Ethernet

Addressing, Switching, and ARP

EC 441 - Introduction to Computer Networking
Boston University

Spring 2026

Learning Objectives

By the end of this lecture, you should be able to:

- Describe the structure of an Ethernet frame and the purpose of each field
- Explain how MAC addresses are structured (OUI, unicast/multicast/broadcast)
- Trace an ARP exchange: how an IP address is resolved to a MAC address
- Explain how an Ethernet switch self-learns its forwarding table
- Trace frame delivery across a multi-switch LAN
- Define a subnet and relate it to ARP and routing
- Identify major Ethernet generations and their speeds

Key Acronyms

Acronym	Meaning
ARP	Address Resolution Protocol
BIA	Burned-In Address
CAM	Content-Addressable Memory
CIDR	Classless Inter-Domain Routing
CRC	Cyclic Redundancy Check
FCS	Frame Check Sequence
IFG	Interframe Gap
IP	Internet Protocol

Acronym	Meaning
LAN	Local Area Network
MAC	Medium Access Control
MLT	Multi-Level Transmission
MTU	Maximum Transmission Unit
NDP	Neighbor Discovery Protocol
NIC	Network Interface Card
OUI	Organizationally Unique Identifier
PAM	Pulse Amplitude Modulation
SFD	Start Frame Delimiter
STP	Spanning Tree Protocol
TTL	Time to Live
UTP	Unshielded Twisted Pair

Ethernet: A Brief History

Origin: Bob Metcalfe & David Boggs, Xerox PARC, 1973 — 2.94 Mb/s over coaxial cable

Timeline:

- 1983: IEEE 802.3 standard ratified (10 Mb/s)
- 1995: Fast Ethernet (100 Mb/s)
- 1999: Gigabit Ethernet (1 Gb/s) — today's desktop standard
- 2002: 10 Gb/s fiber; 2010: 100 Gb/s; 2017: 400 Gb/s

Topology evolution:

- Coaxial bus (all hosts share one wire) → hub → **switch**
- Modern: each host has a dedicated full-duplex link to the switch
- Same frame format throughout — huge backward-compatibility win

[Wikipedia: Ethernet]

Ethernet at the Physical Layer

From lectures 3–4: bits become signals via a **line code** (encoding scheme)

A line code must provide:

- **Clock recovery**: receiver must be able to stay synchronized with the sender — without a separate clock wire
- **DC balance**: no average voltage offset (transformers and AC-coupled links can't pass DC)
- **Bandwidth efficiency**: more bits per unit bandwidth = higher possible data rates

NRZ (Non-Return-to-Zero): 1 = high, 0 = low — simple but fails clock recovery for long runs of identical bits

The next slide shows **Manchester encoding**, used in all 10 Mb/s Ethernet.
Faster generations use more bandwidth-efficient codes (see table after).

Ethernet Line Codes by Generation

Generation	Line code	Notes
10 Mb/s (10BASE-T/5/2)	**Manchester**	Self-clocking; 2× bandwidth cost
100 Mb/s (100BASE-TX)	4B5B + MLT-3	4B5B maps 4 data bits → 5-bit code words; MLT-3 uses 3 voltage levels to cut bandwidth by 3×
1 Gb/s (1000BASE-T)	PAM-5	5-level amplitude modulation on all 4 wire pairs simultaneously
10+ Gb/s	PAM-4 / DSP	4-level PAM with digital signal processing and equalization

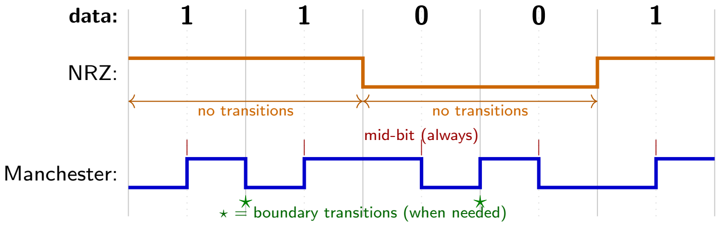
Trend: as speed increases, line codes become more spectrally efficient (more bits per Hz) to stay within cable/fiber bandwidth limits.

[Wikipedia: Manchester code]

Manchester Encoding — 10 Mb/s Ethernet

Rule: every bit period has a transition at its **midpoint**

- Bit **1** $\rightarrow$ low first half, **high** second half (rising edge at center)
- Bit **0** $\rightarrow$ high first half, **low** second half (falling edge at center)



Properties: guaranteed transition every bit period $\rightarrow$ clock recovery always possible; DC-balanced

Cost: signal bandwidth $= 2\times$ bit rate (20 MHz for 10 Mb/s) — why later Ethernet uses other codes

Preamble connection: 10101010$\times$7 in Manchester = perfectly regular 10 MHz square wave for PLL lock

Ethernet Frame Structure

Preamble+SFD	Dst MAC	Src MAC	Type	Payload	FCS
8 B	6 B	6 B	2 B	46–1500 B	4 B

- **Preamble (7 B)**: `10101010`×7 — clock sync for receiver PLL
- **SFD (1 B)**: `10101011` — two final 1-bits mark "data starts now"
- **Dst/Src MAC (6 B each)**: 48-bit link-layer addresses
- **Type (2 B)**: encapsulated protocol (`0x0800`=IPv4, `0x0806`=ARP, `0x86DD`=IPv6)
- **Payload**: 46–1500 B; pad to 46 B if shorter
- **FCS (4 B)**: CRC-32; bad frames silently dropped; not passed to higher layers

Minimum and Maximum Frame Size

Maximum frame size: payload $\leq$ 1500 bytes $\Rightarrow$ **1518 bytes total**

- This is where IP's **MTU of 1500 bytes** comes from

Minimum frame size: payload $\geq$ 46 bytes $\Rightarrow$ **64 bytes total**

- Required for CSMA/CD: sender must still be transmitting when the collision echo arrives
- At 10 Mb/s, max cable length gives round-trip delay $\tau = 51.2\,\mu$s:

$$L_{\text{min}} = 2\tau R = 51.2\,\mu\text{s} \times 10\,\text{Mb/s} = 512\,\text{bits} = \textbf{64 bytes}$$

- In modern full-duplex switched Ethernet, collisions can't occur — 64-byte minimum is a legacy constraint
- If payload $<$ 46 bytes, the sender adds **pad bytes** to reach the minimum
- **IFG** (Interframe Gap): 96-bit silence *after* each frame — this is how the receiver knows the frame has ended

MAC Addresses

A **MAC address** (also called a hardware address or Burned-In Address, **BIA**) is a 48-bit identifier for a network interface, written in hex:

`A4:C3:F0:85:AC:2D`

Structure:

- Bytes 1–3: **OUI** (Organizationally Unique Identifier)
  - Assigned to manufacturers by IEEE
  - `A4:C3:F0` = Apple
- Bytes 4–6: device-specific (assigned by manufacturer)

Special bits in byte 1:

- Bit 0 (I/G): 0 = unicast, 1 = multicast
- Bit 1 (U/L): 0 = globally administered, 1 = locally administered

Broadcast: `FF:FF:FF:FF:FF:FF` (all devices on LAN receive it)

MAC addresses are flat: no topological structure (unlike IP)

[Wikipedia: MAC address]

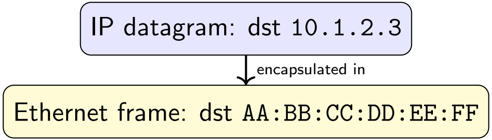
Flat vs. Hierarchical Addressing

MAC (flat)

- Identifies the hardware
- No location information
- Does not change as a device moves (physically)
- Used by switches for **local** frame delivery

IP (hierarchical)

- Encodes network topology in the prefix
- Changes when a device moves between subnets
- Used by routers for **global** routing

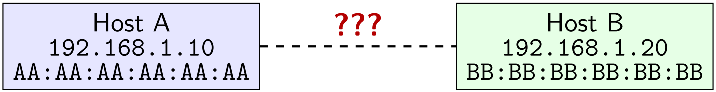


Key: the MAC address changes at *each hop*; the IP address stays the same end-to-end

ARP: The Problem

Host A knows it needs to reach `192.168.1.20` on the same subnet.

To send an Ethernet frame, it needs the **destination MAC address**. Where does it come from?



ARP (Address Resolution Protocol): "Who has IP `192.168.1.20`? Tell me your MAC."

[Wikipedia: Address Resolution Protocol]

ARP Exchange

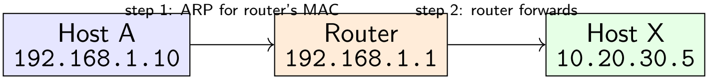
1. **ARP Request** (broadcast):
   - Ethernet dst: `FF:FF:FF:FF:FF:FF` (every host on LAN receives this)
   - Content: "I'm `192.168.1.10` (`AA:AA:AA:AA:AA:AA`). Who has `192.168.1.20`?"
2. **ARP Reply** (unicast):
   - Ethernet dst: `AA:AA:AA:AA:AA:AA` (back to Host A only)
   - Content: "I'm `192.168.1.20`; my MAC is `BB:BB:BB:BB:BB:BB`."
3. **Host A sends the datagram**:
   - Ethernet frame dst: `BB:BB:BB:BB:BB:BB`

ARP table (cache): Host A stores the IP $\rightarrow$ MAC mapping with a TTL ($\sim$20 min). Next time: no ARP needed.

Hosts can also **learn** from ARP requests they overhear — the sender's IP/MAC is in every request.

ARP for Off-Subnet Destinations

Host A (192.168.1.10) wants to reach Host X (10.20.30.5) — different subnet.



1. Host A sees that 10.20.30.5 is **not on the local subnet**
2. Host A consults its routing table → next hop is default gateway 192.168.1.1
3. Host A ARPs for the **gateway's** MAC address
4. Frame is sent to router's MAC; IP datagram inside still has dst 10.20.30.5

ARP operates within one subnet (broadcast domain) only.

Hub vs. Switch

Hub (repeater)

- Receives on one port, retransmits on **all** others
- All ports share one collision domain
- Half-duplex — CSMA/CD required
- Every host sees every frame

Switch

- Forwards frames **selectively** by MAC address
- Each port is its own collision domain
- Full-duplex — no collisions possible
- Frames go only where they need to

Why switches replaced hubs:

- **Performance**: multiple simultaneous transmissions (A→B and C→D at the same time)
- **Privacy**: hosts don't receive frames not addressed to them
- **Full-duplex**: no CSMA/CD needed; full link bandwidth in both directions

[Wikipedia: Network switch]

Switch Forwarding and Self-Learning

Forwarding table (also: MAC address table, CAM table):

MAC Address	Port	TTL
`AA:AA:AA:AA:AA:AA`	1	4:58
`BB:BB:BB:BB:BB:BB`	3	4:45
`CC:CC:CC:CC:CC:CC`	2	4:12

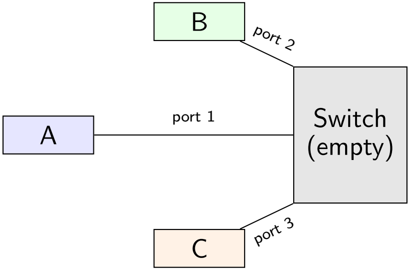
Self-learning: when a frame arrives on port P from source MAC X, record $(X \to P)$.

Forwarding rule:

1. Look up **destination MAC** in table
2. If found $\to$ forward to that port only
3. If not found $\to$ **flood** (all ports except ingress)
4. Broadcast/multicast $\to$ flood always

Entries expire after $\sim$300 s of inactivity (hosts that move get relearned automatically).

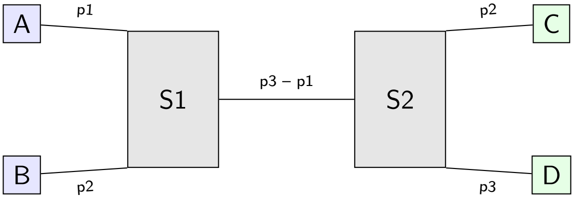
Self-Learning Example



Frame	Switch action	Table after
A $\to$ B (dst unknown)	Learn A$\to$port 1; **flood** ports 2, 3	A$\to$1
B $\to$ A (dst known!)	Learn B$\to$port 2; **forward** to port 1	A$\to$1, B$\to$2
C $\to$ A (dst known!)	Learn C$\to$port 3; **forward** to port 1	A$\to$1, B$\to$2, C$\to$3

After a few exchanges, the switch has **learned all host locations** and forwards efficiently.

Multi-Switch LAN



Scenario: A sends to D (tables empty)

1. S1: learns A→p1; D unknown → flood p2 (to B) and p3 (toward S2)
2. S2: learns A→p1 (via S1); D unknown → flood p2 (C) and p3 (D)
3. D receives; B and C discard

D replies to A (S2 now knows A; S1 now knows D)

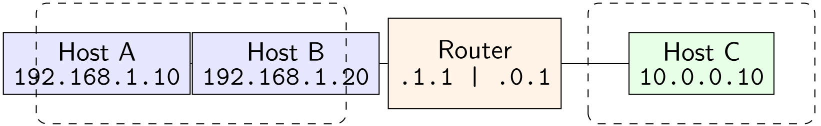
1. S2: D→p3 learned; A known at p1 → **forward** only to S1
2. S1: A known at p1 → **forward** only to A

Switches are **transparent** to end hosts.

Subnets

A **subnet** is a set of interfaces sharing the same IP prefix — reachable at the link layer without a router.

CIDR notation: `192.168.1.0/24`
⇒ first 24 bits are the network prefix; last 8 bits identify hosts (0–255)



- ARP operates **within** a subnet (broadcast domain)
- Cross-subnet traffic goes through a **router** (ARP for router's MAC)
- Each router interface belongs to a different subnet

Ethernet Technology Survey

Naming: `<speed>BASE<medium>` — speed in Mb/s or Gb/s; BASE = baseband

Historical (bus/hub era):

Standard	Year	Speed	Medium
10BASE-5 "Thicknet"	1980	10 Mb/s	RG-8 coax, bus
10BASE-2 "Thinnet"	1985	10 Mb/s	RG-58 coax, bus
10BASE-T	1990	10 Mb/s	Cat 3 UTP, star hub

Modern (switched):

Standard	Year	Speed	Medium
100BASE-TX (Fast Ethernet)	1995	100 Mb/s	Cat 5 UTP
1000BASE-T (Gigabit)	1999	1 Gb/s	Cat 5e UTP
10GBASE-T	2006	10 Gb/s	Cat 6A UTP
100GBASE-SR4/LR4	2010	100 Gb/s	Fiber
400GBASE-SR8	2017	400 Gb/s	Fiber

[Wikipedia: IEEE 802.3]

Power over Ethernet and Auto-Negotiation

Power over Ethernet (PoE) — IEEE 802.3af/at/bt:

- Delivers DC power over the same UTP cable as data
- 802.3af: 15.4 W per port   802.3at: 30 W   802.3bt: up to 90 W
- Used for: IP phones, wireless access points, IP cameras, IoT sensors

Auto-negotiation:

- Interfaces advertise their capabilities; both ends select highest common mode
- 1000BASE-T interfaces can fall back to 100 or 10 Mb/s if needed

Duplex mismatch (common misconfiguration):

- One end full-duplex, other end half-duplex
- Full-duplex end never backs off; half-duplex end sees constant collisions
- Symptom: poor throughput, high CRC error counts

[Wikipedia: Power over Ethernet]

Summary

- **Ethernet frame**: preamble + dst MAC + src MAC + type + payload (46–1500 B) + FCS
  - 64-byte minimum $\leftarrow$ CSMA/CD requirement; 1518-byte maximum $\leftarrow$ 1500-byte MTU
- **MAC address**: 48-bit flat identifier; OUI identifies manufacturer; broadcast = `FF:FF:FF:FF:FF:FF`
- **ARP**: resolves IP $\rightarrow$ MAC within a subnet; request is broadcast, reply is unicast; results cached with TTL
- **Switch**: self-learning forwarding table; forward if known, flood if unknown; full-duplex, no collisions
- **Subnet**: IP prefix defines a broadcast domain; ARP works within it; routing needed to cross subnets
- **Ethernet speeds**: 10 Mb/s $\rightarrow$ 100 Mb/s $\rightarrow$ 1 Gb/s $\rightarrow$ 10/100/400 Gb/s; coax bus $\rightarrow$ switched star

Next: Wireless Networks (802.11) — same frame-layer concepts, different MAC and physical layer